

1. Scope, architecture and output

Astra local property reconstruction — 3 October 2026. 9/12 declared runs have completed at report generation. This is a measured development submission; no blanket gate-pass claim is made.

Route 2 uses Stray Scanner 1.4 on iPhone 15 Pro Max. Laptop: Ubuntu 24, Intel i7-1265U, 64 GB RAM, no GPU. Photos/video are intended for iPhone 15 or newer; other physical devices have not been tested. Kaggle is optional and unvalidated.

LiDAR: validate samples/units -> backproject confident depth -> verified local/loop correction graph -> structural room planes. RGB: select frames/photos -> focal/depth prior -> feature matching and PnP/essential geometry -> relative pose graph -> approximate gravity -> structural layout.

All tiers: surface IDs -> opening/region candidates -> metric projection -> transparent inspection rules and scope -> provisional JSON, room/surface drawings and whole-property plan. Input and model hashes, source provenance, timing and dependency versions accompany each run.

Ground truth is read only by evaluation. The official Round 1 schema is missing; the supplied schema is explicitly provisional. Filtered point fusion and GrabCut replace the planned TSDF and SAM components.

2. Sensor handling, geometry and drift

Depth is millimetres; poses and output coordinates are metres. Per-frame intrinsics are resized to depth resolution. Stray camera-to-world xyzw poses use OpenCV camera coordinates, with world y up. The plan is x,z after a common yaw alignment.

MP4 sample times preserve variable timing. Decoding disables edit-list trimming to retain recorded HEVC sensor frames. Pose/RGB timing residuals are exported. IMU acceleration norms are near 1, so raw translation is not integrated under a false m/s^2 assumption.

Local and revisited cloud pairs create trimmed ICP factors only after overlap/residual/motion checks. A robust sparse small-angle correction graph anchors the first camera and penalizes implausible/non-smooth corrections. It is an approximation, not complete SLAM.

The actual 140-frame ablation accepted 41 factors including five loop factors, with maximum translation correction about 8.89 cm. Summed inferred room area changed from 20.8086 to 20.8020 m^2 . The footprint overlay also exposes shape changes. A small area delta does not prove accuracy improvement.

Distance-watershed free-space basins seed structural-plane extraction. High wall observations reduce furniture confusion; local horizontal modes estimate height. Unobserved ceilings remain null. Low coverage, reflective/glass surfaces, cabinets, soffits and non-Manhattan walls can still bias boundaries.

3. RGB models, semantics and scope

The initial RGB depth prior is Depth Anything V2 Metric Hypersim Small. It executes on CPU but has severe scale bias in the supplied stills. Depth Pro estimates focal length and depth jointly; its actual after-run results are reported separately when complete. No API key or paid inference is used.

Full-precision Depth Pro took about 225.6 seconds including load in a concurrent CPU probe. Dynamic int8 took about 230.5 seconds and changed focal length from 236.6 to 462.2 px on the same input. That variant is excluded from the benchmark configuration; numerical equivalence is not assumed.

Depth-assisted PnP handles some low-parallax feature pairs that defeat essential triangulation. Remaining disconnected components are explicitly schematic. Overlap and adjacency diagnostics must pass before a physical whole-property stitch can be accepted.

Grounding DINO proposals and GrabCut masks are unverified visual evidence. LiDAR openings additionally require wall gaps plus transmitted depth rays and lintel support. Facing opening pairs support room adjacency; detected widths remain uncertain.

Black/brown staging props are assessed only in explicit staged mode. They are not natural damage training examples. Metric wall regions are clipped and unioned; generalized floor/ceiling damage is incomplete. Flags recommend inspection using named rules. Scope items are inspection quantities, not diagnoses or priced repair orders.

4. Benchmark, gates and uncertainty

Dataset: kitchen, hall and corridor; furnished staged-damage capture; three evaluator sample scans. All supplied data are retained. This is two rooms plus a corridor, short of the required three rooms plus connector. Repeat scans are deferred; consumer-app exports are unavailable tonight.

Laser truth contains room length/breadth/height and room-level doorway entries. Wall identities, exhaustive opening IDs, global outline and damage reference extents are absent. Sorted extents are therefore proxies, not full per-wall scores. Video physical correspondence can remain unresolved and is not chosen to minimize error.

See benchmark.md and metrics.csv for every measured row, including missing predictions. The early LiDAR after run estimated hall height 2.7687 m, kitchen 2.7591 m and corridor 2.1928 m against 2.8, 2.8 and 2.26 m. All miss the 1.5 cm gate. Later results are tabulated separately.

Every measurement includes a nominal 95% engineering interval, explicitly marked uncalibrated. Unknown values and intervals are null. Detector scores and sensor confidence are not calibrated geometric intervals. Nominal coverage on this small correlated set is descriptive only.

Eight kitchen stills are duplicated across folders. No independent calibration/test split is claimed. Calibrated coverage requires separate properties, complete measurement correspondence, reference uncertainty and held-out validation. Missing repeatability/app/damage evidence is not a pass.

5. Fix loop and engineering evidence

The first declaration targeted the worst measured LiDAR height error available at that time: kitchen error 4.65 cm. It hypothesized that global floor support and free-space partitioning contaminated room measurements. It predicted at most 2 cm kitchen/hall error after local structural-plane extraction.

The shipped plane change substantially improves the kitchen/corridor boundaries and enables ray-supported adjacency. It misses the predicted height improvement; corridor height regresses. This supports the wall-partition diagnosis but leaves sensor/pose/surface-selection height bias unresolved. No laser scale fitting was used.

After the first complete photo baseline, a separate declaration identified a worse gate: corridor long-extent error about 232.9%, six disconnected components and no adjacency. The proposed fix changes focal/depth, pose estimation and gravity; the prediction is worst extent error below 100% plus one observed connection, still short of full compliance.

Before outputs, declarations, evaluation tables and readable code diff are retained in `fix_loop`. The reproduction script can regenerate the methods. The photo prediction is judged only when a completed after run exists. Failures are retained instead of replaced by a favourable subset.

Development commits use `.history` because this coding session mounts `.git` read-only. A standard Git bundle exports the actual incremental history. Unit/integration checks cover projection units, rigid alignment, gravity, masks, doorway evidence, topology and reference validation. Accuracy still requires physical benchmark evidence.

6. Reproduction, defense and remaining risks

README supplies one command per tier/capture. setup.sh installs pinned packages and fetch scripts download public models; inference then runs offline. A fresh-machine setup under 15 minutes has not been measured. Download bandwidth and Depth Pro CPU inference are substantial costs.

The allowlisted reproduction archive contains source, declarations, raw data, measurements, outputs, manifests, history and optionally model weights. It excludes credentials and virtual environments. Exact-content caches speed replay while the live model path remains executable. Timings distinguish observed executions and cache reuse through provenance.

The stock capture page asks for slow overlapping views of floor/ceiling junctions, both sides of doorways and a return loop. The photos need parallax and shared doorway detail. Mirrors/glass, glossy or wet-looking finishes, motion blur and low light remain known failure modes, rather than solved claims.

Before a compliant submission: obtain official contract/schema, collect complete opening/wall/damage reference correspondence and independent calibration scenes, measure repeatability, acquire the two-room app exports, and test the capture instructions on an unseen property. The user has deferred physical additions for this iteration.

The delivered output must be read with its warnings: LiDAR geometry is useful but centimetre gates fail; RGB scale/stitching may fail; staged masks do not validate natural damage; concealed flags are inspection prompts. docs/COMPLIANCE.md maps every requirement to its artifact and actual status.